

1.

A rectangular channel 1.5 m wide has a discharge of 200 litres per second, which is measured by a right-angled V-notch weir. Find the position of the apex of the notch from the bed of the channel if maximum depth of water is not be exceed 1 m. Take $C_d = 0.62$. [Ans. .549 m]

$$Q = 0.2 \text{ m}^3/\text{s}$$

$$C_d = 0.62$$

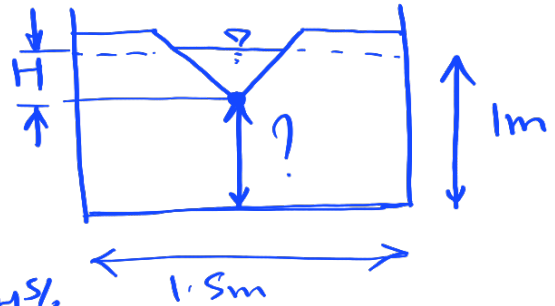
$$Q = \frac{8}{15} C_d \tan \frac{\theta}{2} \sqrt{2g} H^{5/2}$$

$$= \frac{8}{15} \times 0.62 \times \tan 45^\circ \sqrt{2g} \times H^{5/2}$$

$$\Rightarrow H = 0.451 \text{ m} \Rightarrow 1 - H = 0.549 \text{ m}$$

$$\frac{V_a^2}{2g} = \frac{\left(\frac{0.2}{1.5 \times 1}\right)^2}{2g} = \frac{(0.133)^2}{2g} = 0.0009 \text{ m}$$

↓
negligible on
compared to
H



2.

Find the time required to lower the water level from 3 m to 1.5 m in a reservoir of dimension $70 \text{ m} \times 70 \text{ m}$, by a rectangular notch of length 2.0 m. Take $C_d = 0.60$. [Ans. 11 min 1 s]

$$\begin{aligned} T &= \frac{3A}{C_d L \sqrt{2g}} \left[\frac{1}{\sqrt{H_2}} - \frac{1}{\sqrt{H_1}} \right] \\ &= \frac{3 \times 70 \times 70}{0.6 \times 2 \times \sqrt{2 \times 9.81}} \left[\frac{1}{\sqrt{1.5}} - \frac{1}{\sqrt{3}} \right] \\ &= \underline{661.37 \text{ s.}} \end{aligned}$$

3.

Water is flowing in a rectangular channel of 1.2 m wide and 0.8 m deep. Find the discharge over a rectangular weir of crest length 70 cm if the head of water over the crest of weir is 25 cm and water from channel flows over the weir. Take $C_d = 0.60$. Neglect end contractions but consider velocity of approach.

[Ans. $0.1557 \text{ m}^3/\text{s}$]

$$Q = \frac{2}{3} C_d L \sqrt{2g} \left[(H + h_a)^{3/2} - h_a^{3/2} \right]$$

$$h_a = \frac{v_a^2}{2g}$$

with $v_a = 0 \text{ m/s}$; find Q

$$Q_1 = \frac{2}{3} \times 0.6 \times 0.7 \times \sqrt{2g} \times \left[0.25^{3/2} \right] = 0.1547 \text{ m}^3/\text{s}$$

$$v_a = \frac{0.1547}{1.2 \times 0.8} = 0.161 \text{ m/s}$$

$$h_a = \frac{v_a^2}{2g} = 0.0013 \text{ m}$$

$$Q_2 = \frac{2}{3} \times 0.6 \times 0.7 \times \sqrt{2g} \times \left[(0.25 + 0.0013)^{3/2} - 0.0013^{3/2} \right] = 0.156 \text{ m}^3/\text{s}$$

...

Q_3

...

Q_4

\

4.

A discharge of $1500 \text{ m}^3/\text{s}$ is to pass over a rectangular weir. The weir is divided into a number of openings each of span 7.5 m . If the velocity of approach is 3 m/s , find the number of openings needed in order the head of water over the crest is not to exceed 1.8 m [Ans. 37.5 say 38]

$$Q = N 1.84 (L - 0.1 n H_1) (H_1^{3/2} - h_a^{3/2})$$

$$h_a = \frac{v_a^2}{2g} = \frac{3^2}{2 \times 9.81} \text{ m}$$

$$H_1 = H + h_a = 1.8 + \frac{3^2}{2 \times 9.81}$$

$$\Rightarrow 1500 = N \times 1.84 (7.5 - 0.1 \times 2 \times H_1) (H_1^{3/2} - h_a^{3/2})$$

$$\Rightarrow N \approx 38$$

5.

The head of water over a rectangular weir is 50 cm. The length of the crest of the weir with end contraction suppressed is 1.4 m. Find the discharge using following formulae : (i) Francis's Formula and (ii) Bazin's Formula.
[Ans. (i) $0.91 \text{ m}^3/\text{s}$, (ii) $.901 \text{ m}^3/\text{s}$]

$$Q = 1.84 (L - 0.1 n H_1) (H_1^{3/2} - h_a^{3/2})$$

$$= 1.84 (1.4) \times (0.5^{3/2}) = 0.91 \text{ m}^3/\text{s}.$$

↳ Francis.

Bazin

$$Q = m_1 \sqrt{2g} L H_1^{3/2} = \left(0.405 + \frac{0.003}{0.5} \right) \times \sqrt{2g} \times 1.4 \times 0.5^{3/2}$$

$$m_1 = 0.405 + \frac{0.003}{H_1}$$

$$H_1 = H + \alpha \frac{V_a^2}{2g}$$

↳ 1.0

$$= 0.901 \text{ m}^3/\text{s}$$

6.

Find the discharge over a cipolletti weir of length 1.8 m when the head over the weir is 1.2 m. Take $C_d = 0.62$ [Ans. 4.331 m³/s]

$$\begin{aligned} Q &= \frac{2}{3} C_d L \times \sqrt{2g} \times H^{3/2} \\ &= \frac{2}{3} \times 0.62 \times 1.8 \times \sqrt{2g} \times 1.2^{3/2} \\ &= 4.332 \text{ m}^3/\text{s}. \end{aligned}$$

7.

- (a) A broad-crested weir of length 40 m, has 400 mm height of water above its crest. Find the maximum discharge. Take $C_d = 0.6$. Neglect velocity of approach. [Ans. 10.352 m³/s]
 (b) If the velocity of approach is to be taken into consideration, find the maximum discharge when the channel has a cross-sectional area of 40 m² on the upstream side. [Ans. 10.475 m³/s]

$$\begin{aligned}
 Q &= C_d L \times \frac{2}{3} H \times \sqrt{2g} \times \left(\frac{H}{3}\right)^{1/2} \\
 &= 0.6 \times 40 \times \frac{2}{3} \times 0.4 \times \sqrt{2g} \times \left(\frac{0.4}{3}\right)^{1/2} \\
 &= 10.351 \text{ m}^3/\text{s}. \rightarrow \text{without } \underline{v_a}
 \end{aligned}$$

$$\begin{aligned}
 Q &= C_d \times L \times \frac{2}{3} \times \frac{\sqrt{2g}}{\sqrt{3}} \times (H_1)^{3/2} \\
 &= 0.6 \times 40 \times \frac{2}{3} \times \frac{\sqrt{2g}}{\sqrt{3}} \times (0.403)^{3/2} = 10.48 \text{ m}^3/\text{s}
 \end{aligned}$$

$$\begin{aligned}
 H_1 &= H + \frac{v_a^2}{2g} \\
 &= 0.4 + \frac{\left(\frac{10.351}{40}\right)^2}{2g} = \underline{\underline{0.403 \text{ m}}}
 \end{aligned}$$

8.

The heights of water on the upstream and downstream side of a sub-merged weir of length 3.5 m are 300 mm and 150 mm respectively. If C_d for free and drowned portion are 0.6 and 0.8 respectively, find the discharge over the weir. [Ans. 1.0807 m³/s]

$$Q_1 = \frac{2}{3} C_{d1} \sqrt{2g} L \times (H_1 - H_2)^{3/2}$$

Freely discharging weir

$$= \frac{2}{3} \times 0.6 \times \sqrt{2g} \times 3.5 \times (0.3 - 0.15)^{3/2}$$

$$= 0.36 \text{ m}^3/\text{s}$$

$$Q_2 = C_{d2} L \times H_2 \times \sqrt{2g(H_1 - H_2)}$$

Drowned orifice.

$$= 0.8 \times 3.5 \times 0.15 \times \sqrt{2g(0.15)}$$

$$= 0.72 \text{ m}^3/\text{s}$$

$$Q = Q_1 + Q_2 = \underline{1.08 \text{ m}^3/\text{s}}$$

